

MENOMONIE, WI, USA

WMO: 720327

Lat: 44.892N

Lon: 91.868W

Elev: 895

StdP: 14.23

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04995

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.2	-7.6	-20.0	1.9	-10.9	-16.5	2.3	-6.2	24.0	17.6	20.8	15.8	5.3	270	0.549

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	89.6	71.6	86.0	70.5	82.5	68.3	75.3	84.2	73.3	81.6	71.6	79.7	11.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.6	124.8	80.0	70.8	117.1	78.0	69.3	111.4	76.5	39.4	84.8	37.5	81.1	36.0	80.0	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.9	18.7	16.8	DB	-18.6	93.5	5.7	4.3	-22.7	96.6	-26.0	99.1	-29.2	101.5	-33.3	104.6	(4)
			WB	-19.0	78.5	5.6	1.9	-23.0	79.8	-26.3	81.0	-29.5	82.0	-33.5	83.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.4	15.9	18.6	32.3	45.5	57.9	67.1	71.5	68.8	61.7	48.2	34.5	20.9
	DBStd	21.77	12.69	11.90	12.32	9.81	8.77	6.03	5.38	5.71	7.75	8.92	10.06	11.23
	HDD50	4246	1059	878	561	198	26	0	0	0	7	141	473	903
	HDD65	7716	1524	1298	1013	585	252	46	9	27	153	525	916	1368
	CDD50	2561	0	0	14	63	272	513	667	582	357	86	7	0
	CDD65	557	0	0	0	1	33	110	211	144	53	5	0	0
	CDH74	4959	0	0	7	36	381	957	1911	1162	455	50	0	0
	CDH80	1402	0	0	0	4	112	248	629	303	101	5	0	0
Wind	WSAvg	7.1	7.3	7.4	7.3	8.6	7.9	6.9	5.9	5.6	6.4	7.3	7.5	7.2
Precipitation	PrecAvg	33.6	1.1	1.1	2.1	3.0	3.9	5.3	4.0	4.2	3.5	2.4	1.9	1.2
	PrecMax	41.2	2.6	2.2	5.2	6.3	7.0	9.0	9.1	8.9	8.9	5.7	7.2	2.4
	PrecMin	26.8	0.2	0.4	0.4	0.8	1.8	2.4	1.0	0.4	0.5	0.6	0.0	0.3
	PrecStd	4.3	0.7	0.5	1.1	1.5	1.6	1.8	1.7	2.3	2.0	1.6	1.7	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.9	51.0	72.4	79.1	89.7	90.4	95.0	90.6	88.0	79.9	67.1	46.7
		MCWB	38.3	46.1	59.2	59.1	67.1	72.6	72.7	72.0	71.7	64.4	55.6	41.1
	2%	DB	38.7	42.5	62.9	72.1	82.2	86.2	90.1	86.4	82.3	73.4	59.2	42.0
		MCWB	35.0	37.9	53.3	55.1	64.1	69.4	72.9	71.3	68.7	62.2	51.2	38.8
	5%	DB	35.9	37.4	55.2	67.5	78.5	82.1	86.2	82.9	79.4	69.2	54.0	37.2
		MCWB	33.9	33.8	47.6	52.5	62.5	67.1	71.3	69.2	66.7	59.2	46.6	35.1
	10%	DB	33.3	34.3	48.3	62.9	73.0	79.5	83.0	80.9	75.5	63.0	49.4	35.3
		MCWB	31.6	31.7	42.7	49.8	59.1	65.8	69.5	68.0	65.9	54.6	44.4	33.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.9	44.5	61.2	61.1	72.6	76.8	78.8	76.3	74.7	70.7	57.6	45.1
		MCDB	43.7	50.6	72.5	72.7	84.5	85.9	87.5	85.0	83.1	74.4	62.2	45.5
	2%	WB	35.6	38.4	55.5	57.9	68.7	72.8	76.2	74.2	71.9	64.1	52.4	39.1
		MCDB	37.9	42.2	59.0	68.2	77.4	81.2	85.4	82.1	78.9	69.8	59.4	41.4
	5%	WB	33.7	34.5	48.0	54.5	65.6	70.5	73.8	72.4	69.4	59.7	47.6	35.8
		MCDB	35.4	37.3	56.0	63.6	73.0	78.8	82.6	79.9	75.7	66.6	52.8	37.5
	10%	WB	31.5	32.0	43.1	51.3	62.4	68.4	71.7	70.6	67.3	55.6	44.1	33.6
		MCDB	32.7	33.8	48.1	60.8	69.9	76.4	80.4	77.5	74.1	61.8	48.7	34.5
Mean Daily Temperature Range	5% DB	MDBR	15.5	16.8	17.2	19.5	20.8	19.4	19.8	19.7	20.4	18.2	14.8	13.2
		MCDBR	16.3	17.6	24.8	28.9	27.2	23.2	22.9	22.7	24.6	26.5	23.2	15.1
		MCWBR	13.9	13.6	15.3	15.2	12.4	10.5	9.4	10.1	12.1	14.8	15.2	12.7
	5% WB	MCDBR	15.7	16.5	22.8	25.3	21.7	19.9	19.9	19.9	19.9	21.5	21.0	13.5
		MCWBR	13.8	13.1	15.2	15.3	12.1	10.9	10.6	10.1	11.1	14.1	14.8	11.6
Clear-Sky Solar Irradiance	taub	0.257	0.285	0.318	0.358	0.393	0.394	0.382	0.391	0.362	0.322	0.289	0.264	
	taud	2.394	2.306	2.355	2.342	2.286	2.319	2.368	2.354	2.414	2.510	2.508	2.434	
	Ebn at Noon	278	286	291	286	278	278	279	272	272	271	264	263	
	Edh at Noon	23	30	34	37	41	40	37	36	32	25	21	20	
All-Sky Solar Radiation	RadAvg	546	893	1234	1448	1688	1873	1964	1647	1289	819	516	381	
	RadStd	58	85	121	150	132	146	75	111	87	104	49	41	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page